Hierarchy beats the matched transformer: 16–50% lower cross-entropy, 5,430× less state

Equal-storage comparison at 10.00M trained tokens · 119.7M parameters each · distilled from Qwen3-0.6B-Base · trained at 512 tokens, evaluated at 512 / 1,024 / 2,048

THESIS: A GEOMETRIC APPROACH TO SELF-ATTENTION

Standard self-attention keeps a key/value cache that grows with context and a fixed set of stored weight matrices. We replace both. **Function-weights (FW)** represent weights as functions rather than stored matrices, so the effective weight at each position is computed from the geometry of the input instead of looked up. **Hierarchy** replaces the KV cache with a hierarchical recurrent state — a small set of tensors at multiple scales that summarize the sequence, so state is constant in context. Attention becomes geometry over a compact recurrent representation instead of a dot-product search over every past token.

Function-weights
Weights as functions, not matrices. Width-688 model; +FW variants (local, hierarchy, multiscale) match pure hierarchy within 0.04 CE at every checkpoint.

Hierarchical recurrent state
4 recurrent tensors × 688 × bf16 = 5,504 bytes per sequence, independent of context. 114.1M MACs/token at any length; 7% fewer than the transformer at 512, 15% at 2,048.

16.3% → 50.3%
LOWER FIXED CE, 512 → 2,048
16.3% at 512 · 40.7% at 1,024 · 50.3% at 2,048

5,430×
LESS PERSISTENT STATE AT 2K
5,504 B vs 29.88 MB; 1,357× at 512

+2.27 pt
BENCHMARK MEAN (LIMIT 1,000)
0.280 vs 0.258; top-1 agreement 0.223 → 0.318

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SYNCHRONIZED CHECKPOINTS WON
All variants below transformer at every snapshot; mean gap 0.94 CE

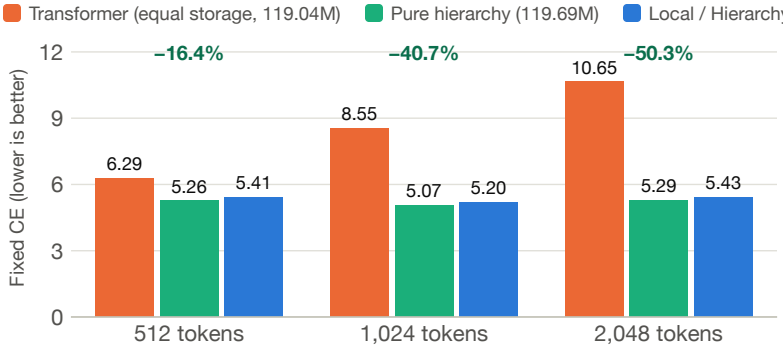


Fig 1. Fixed cross-entropy at 10.00M trained tokens. Green labels: pure-hierarchy reduction vs transformer. Transformer CE grows with context; hierarchy and Local + FW stay flat at 5.1–5.4.

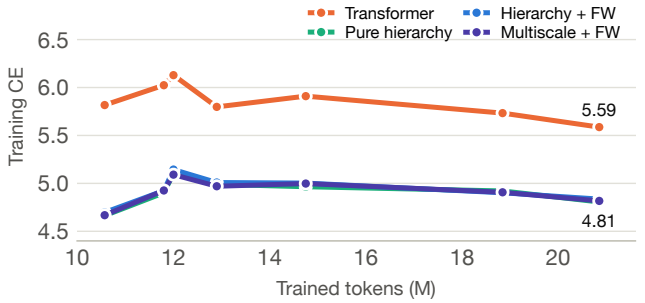


Fig 2. Matched-token training CE, 7 synchronized snapshots (10.6M–20.9M tokens). All three hierarchy variants — with and without function-weights — track together, 0.75–1.15 CE below the transformer at every point.

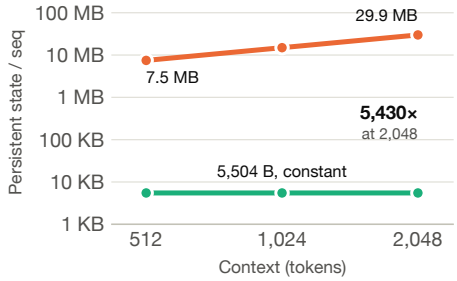


Fig 3. Persistent per-sequence state (log scale). Transformer state grows linearly with context; hierarchy is fixed at 5,504 bytes. Analytic bf16 accounting from resolved configs.

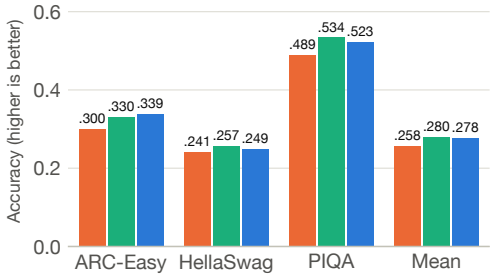


Fig 4. Exact benchmark scores, limit 1,000. Hierarchy leads on ARC-Easy, HellaSwag, PIQA and mean; Local + FW leads ARC-Easy. LAMBADA is 0.000 for all models and omitted from the plot.

EXTERNAL CHECK: OFFICIAL NANOGPT

CONTEXT	NANOGPT	HIERARCHY	Δ
512	5.126	5.260	+2.5%
1,024	5.261	5.070	**−3.6%**
2,048	5.988	5.291	**−11.6%**
KV / state @2K	73.5 MiB	5.4 KiB	14,003×

Same data, teacher, exposure, seeds and eval protocol; nanoGPT pinned to upstream commit 3adf61e, 119.69M params (Δ 0.008%). The stronger external baseline confirms the long-context reversal and the state advantage.

WHAT IS ESTABLISHED

- **Controlled contract.** Byte-identical data, same teacher and 0.1 CE + 1.0 forward-KL objective, same 4,883-step / 10.0M-token exposure, same AdamW schedule and seeds; parameters matched to 0.5% (in-house) and 0.008% (nanoGPT). Artifacts SHA-256 hashed.
- **Long-context reversal holds against both baselines.** Hierarchy fixed CE barely moves from 512 → 2,048 (5.26 → 5.29) while both transformers degrade (in-house 6.29 → 10.65; nanoGPT 5.13 → 5.99).
- **Function-weights preserve quality.** All +FW variants sit within 0.04 CE of pure hierarchy at every checkpoint and within 0.15 at every fixed-eval context, and lead on ARC-Easy.
- **Structurally cheaper.** Constant 5,504-byte state; 114.1M MACs/token at any context vs 122.7–133.9M for the transformer.

SCOPE AND NEXT STEPS

- **Length-extrapolation confound.** Both models trained at 512 and were evaluated at up to 2,048. nanoGPT uses learned absolute position embeddings, so part of its 1K/2K degradation is positional extrapolation; a baseline trained at 2,048 or using RoPE/ALiBi would isolate the architectural effect.
- **Small long-context samples.** 1,024 and 2,048 fixed evals use 2 batches × 1 sequence (512 uses 16 × 2); one seed, no confidence intervals. Treat as strong directional evidence.
- **FW fixed evaluation** is complete for Local + FW; Hierarchy + FW and Multiscale + FW are queued for fixed eval.
- **Scale-up.** 1B-token confirmation run in progress.

All CE values are held-out fixed evaluation unless labeled training CE. State figures are analytic bf16 per-sequence architecture accounting (excludes parameters, activations and allocator overhead). Benchmarks: exact, limit 1,000, one seed.